DEPI — Digital Egypt Pioneers Initiative

Telecom Customer Churn

Analysis & Prediction

Project Documentation

Track: Data Analysis Specialist

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Introduction

1.1 Project Overview

This project presents a comprehensive multi-tool analysis of customer churn behavior within the telecommunications industry. Using a real-world dataset originally published by IBM, the team performed end-to-end data analysis across five platforms: Microsoft Excel, SQL, Python, Power BI, and Tableau — each offering a unique analytical lens on the same underlying data.

The dataset encompasses 7,043 customer records with 38 attributes covering demographics, subscribed services, contract details, billing information, and churn outcomes. A data preprocessing phase involved merging two complementary datasets, cleaning inconsistencies, and structuring the data into a Star Schema data model consisting of one Fact table and four Dimension tables.

1.2 Motivation

Customer churn — the phenomenon where subscribers discontinue their service — is one of the most critical challenges facing the telecommunications industry. Acquiring a new customer costs five to ten times more than retaining an existing one, making churn prediction and prevention a high-priority strategic concern.

26.5% Overall Churn Rate	**$3.68M** Revenue Lost to Churn	**45.8%** Month-to-Month Churn

These figures underscore the business urgency of this project. By applying descriptive, diagnostic, and inferential analytics, the team aims to identify the root causes of churn, quantify the financial impact, and deliver actionable recommendations that can directly reduce churn rates.

1.3 Objectives

This project is designed to achieve the following objectives:

Build a robust data model	Design and implement a Star Schema (Fact + 4 Dimension tables) to enable efficient multi-dimensional analysis.
Perform descriptive analysis	Summarize key metrics including churn rate, average monthly charges, tenure distribution, and revenue breakdown across customer segments.
Identify churn drivers	Determine which contract types, service combinations, demographic groups, and payment methods are most strongly associated with churn.
Apply inferential statistics	Validate findings using T-Tests, Chi-Square tests, ANOVA, and regression analysis to confirm that observed patterns are statistically significant.
Deliver visual insights	Create interactive dashboards in Excel, Power BI, and Tableau that communicate findings clearly to both technical and non-technical stakeholders.
Provide actionable recommendations	Translate analytical findings into concrete business strategies targeting retention, pricing, and customer engagement.

1.4 Challenges

The team encountered and resolved several challenges throughout the project lifecycle:

- Data Integration: Merging two datasets from different sources required careful alignment of column structures, handling of duplicate records, and resolution of naming inconsistencies across 38 attributes.

- Data Model Design: Determining the optimal Star Schema structure involved critical decisions around which attributes belong in the Fact table versus Dimension tables — particularly for customer attributes and churn reasons, which required understanding Kimball methodology trade-offs.
- Relationship Cardinality: Excel Power Pivot enforced 1-to-many constraints that required rethinking the customer dimension approach, leading to a deliberate architectural decision to retain customer attributes in the Fact table.
- Statistical Interpretation: Translating p-values and test statistics from T-Tests, Chi-Square, and ANOVA into plain business language required bridging the gap between statistical rigor and managerial relevance.
- Cross-Tool Consistency: Ensuring that insights derived from Excel, SQL, Python, Power BI, and Tableau remained consistent and complementary across all five platforms presented an ongoing coordination challenge.